PAVAN SRINIVAS DOMA

Machine Learning Engineer
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Machine Learning Engineer with a Master's in Engineering Science and over a year as a data engineer at the University at Buffalo. Skilled in feature engineering, Python and Scikit-learn, with academic projects in credit scoring & customer segmentation and retrieval-augmented conversational AI. Built a RAG system with 97.5% semantic relevance accuracy, containerized on AWS ECS.

SKILLS

Languages: Python, SQL
Machine Learning: Natural Language Processing, Keras, PyTorch, Scikit-Learn, TensorFlow, Feature Engineering, MLflow
LLMs & GenAI: RAG Systems, Fine-tuning, Hugging Face, LLM Evaluation, LangGraph, Prompt Engineering
Data & Analytics: Matplotlib, NumPy, Pandas, Data Quality, Data Analytics
Cloud & DevOps: Docker

PROFESSIONAL EXPERIENCE

Data Engineer - Lead Buffalo, USA
University at Buffalo Sep 2024 - Dec 2025

- Built a production resource allocation pipeline with SQL window functions and time-series decomposition, working with stakeholders to validate the business logic from extraction to the insights shared with leadership.
- Standardized and documented ETL pipelines for stock tracking and allocation, adding data quality gates and error handling that cut data ingestion errors by 30%; wrote runbooks that helped onboard new engineers faster.
- Built real-time KPI dashboards in advanced SQL (window functions, CTEs, aggregations) and optimized the queries behind them, cutting query runtime by 40%.
- Improved ETL pipeline performance through SQL query refactoring and schema redesign, supporting real-time operational decisions that raised asset utilization efficiency by 25%.

ACADEMIC PROJECTS

Credit Scoring & Customer Segmentation

- Developed a credit risk scoring model and defined its feature engineering strategy with a cross-functional team.
- Compared 5 model approaches in MLflow with senior engineers and checked the business logic against data quality trade-offs.

Retrieval-Augmented Conversational AI Systems

- Designed a retrieval-augmented generation (RAG) system and validated its semantic relevance at 97.5% accuracy with ML engineers; containerized it on AWS ECS with monitoring.
- Implemented context engineering, conversation memory and structured output validation so the assistant gave consistent, context-aware answers.

Consumer Complaint Classification

- Built an intent classifier for banking customer queries covering 77 intent classes, reaching 89.5% accuracy.
- Added a confidence threshold that let 49.6% of predictions skip manual review while keeping 99.8% accuracy on the auto-approved cases.

EDUCATION

University at Buffalo, SUNY Buffalo, USA
Master's in Engineering Science Jan 2026

S.R.M Institute of Science and Technology, SRM University Chennai, India
Bachelor of Technology in Artificial Intelligence May 2024

CERTIFICATIONS

- Claude with Amazon Bedrock (Generative AI & LLM Application Development) Jul 2026
- AWS Cloud Practitioner (Cloud platform deployment & AI services) Jan 2026